ClipSAT : Linear Algebra Vectors, Matrices, Eigenvalues & Least Squares [ANSWER KEY]

Unit 8.01: Orthogonality, Gram-Schmidt, QR & Least Squares

Q1. For which k are langle 0, -2, -2 rangle and langle -1, 2, k rangle orthogonal?

[ ] A: 2
[x] B: -2
[ ] C: -1
[ ] D: 3

Solution: Set the dot product to zero: - 2 k - 4=0 , so k = -2 .

Q2. Which set is NOT orthonormal?

[ ] A: {langle frac{5}{13}, frac{12}{13} ra
[x] B: {langle frac{1}{2}, frac{1}{2} rangl
[ ] C: {langle frac{1}{3}, frac{2}{3}, frac
[ ] D: {langle frac{sqrt{2}}{2}, frac{sqrt{

Solution: {langle frac{1}{2}, frac{1}{2} rangle, langle frac{1}{2}, - frac{1}{2}
rangle} fails: either the dot product is nonzero or a vector does not have length 1.
The other sets pass both tests.

Q3. Which matrix is orthogonal satisfies Q^TQ=I ?

[ ] A: left[begin{matrix}1 & 10 & 1end{matr
[ ] B: left[begin{matrix}frac{1}{2} & frac{
[x] C: left[begin{matrix}0 & -11 & 0end{mat
[ ] D: left[begin{matrix}1 & -11 & 1end{mat

Solution: The columns of left[begin{matrix}0 & -11 & 0end{matrix}right] are
orthonormal, so Q^TQ=I . Each other matrix has columns that are not orthogonal or
not unit length.

Q4. Normalize mathbf v=langle 6, 8 rangle .

[x] A: langle frac{3}{5}, frac{4}{5} rangle
[ ] B: langle frac{3}{50}, frac{2}{25} rang
[ ] C: langle frac{3}{7}, frac{4}{7} rangle
[ ] D: langle - frac{3}{5}, - frac{4}{5} ra

Solution: |mathbf v|=10 , so mathbf v/|mathbf v|=langle frac{3}{5}, frac{4}{5}
rangle .

Q5. Which set is orthonormal?

[ ] A: {langle 1, 2, 2 rangle, langle 2, 1,
[x] B: {langle frac{1}{3}, frac{2}{3}, frac
[ ] C: {langle frac{1}{2}, frac{1}{2} rangl
[ ] D: {langle frac{3}{5}, frac{4}{5} rangl

Solution: {langle frac{1}{3}, frac{2}{3}, frac{2}{3} rangle, langle frac{2}{3},
frac{1}{3}, - frac{2}{3} rangle} : the vectors are orthogonal and each has length 1.
Each other set fails one of these tests.

Q6. For which k are langle 3, 3, 2 rangle and langle 2, k, 1 rangle orthogonal?

[ ] A: frac{8}{3}
[ ] B: - frac{5}{3}
[ ] C: frac{11}{3}
[x] D: - frac{8}{3}

Solution: Set the dot product to zero: 3 k + 8=0 , so k = - frac{8}{3} .

Unit 8.01: Orthogonality, Gram-Schmidt, QR & Least Squares

Q7. Normalize mathbf v=langle -2, 1, -2 rangle .

[ ] A: langle - frac{2}{9}, frac{1}{9}, - f

[ ] B: langle - frac{2}{5}, frac{1}{5}, - f

[ ] C: langle frac{2}{3}, - frac{1}{3}, fra

[x] D: langle - frac{2}{3}, frac{1}{3}, - f

Solution: |mathbf v|=3 , so mathbf v/|mathbf v|=langle - frac{2}{3}, frac{1}{3}, - frac{2}{3} rangle .

Q8. For which k are langle 0, -3, 1 rangle and langle 1, k, -1 rangle orthogonal?

[ ] A: frac{1}{3}

[ ] B: frac{2}{3}

[x] C: - frac{1}{3}

[ ] D: frac{4}{3}

Solution: Set the dot product to zero: - 3 k - 1=0 , so k = - frac{1}{3} .